



Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8496**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2023601004
Course Section:	1				

L_29_528_986_L_1505_38403



Name of the candidate. Atharwa Dahgaonkar

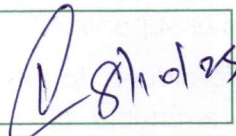
Candidate's UID number:

2	0	2	3	6	0	1	0	0	4
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Examination class room number:

4	0	1
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Invigilator's signature with date:


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(To be filled in by the candidate)

EXAMINATION: BE BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : VII

PROGRAM: AIML - CSF

I / II / III / IV / V / VI VII / VIII

DAY AND DATE: Wednesday, 8th Oct 25

COURSE NAME: Deep learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q1a.

Given:-

$$J(w) = \frac{1}{2} \|x_w - y\|^2$$

Assumption:-

$$f(w) = \frac{1}{1 + e^{-(wx+b)}}$$

To find:-

gradient update rule for gradient descent.

 ~~$\frac{\partial J}{\partial w, b}$~~ ~~$\frac{\partial J}{\partial w}$~~ ~~$\frac{\partial J}{\partial b}$~~ ~~$\frac{\partial J}{\partial w}$~~

Solution.

$$\nabla_w J = \frac{\partial J(w, b)}{\partial w}$$

$$\text{We know that } J(w, b) = \left[\frac{1}{2} \|x_w - y\|^2 \right]$$

$$\nabla_w J = \frac{\partial}{\partial w} \left[\frac{1}{2} \cdot \|x_w - y\|^2 \right]$$

$$\nabla_w J = \frac{1}{2} \cdot 2x_w - y \left[\frac{\partial}{\partial w} f(w) \right]$$

$$\text{Now } f(w) = \frac{1}{1 + e^{-(wx+b)}}$$

$$\nabla_w J = f(x_w) - y$$

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$$D_w t = f(n) - y \frac{\partial}{\partial w} \left[\frac{1}{1 + e^{-(wn+b)}} \right]$$

Solving

$$\frac{\partial}{\partial w} \left[\frac{1}{1 + e^{-(wn+b)}} \right]$$

$$= \frac{-1}{(1 + e^{-(wn+b)})^2} \frac{\partial}{\partial w} [e^{-(wn+b)}]$$

$$= \frac{-1}{(1 + e^{-(wn+b)})^2} \cdot e^{-(wn+b)} \frac{\partial}{\partial w} [-(wn+b)]$$

$$= \frac{-1}{1 + e^{-(wn+b)}} \cdot \frac{e^{-(wn+b)}}{1 + e^{-(wn+b)}} \cdot -n$$

Now we know $\frac{1}{1 + e^{-(wn+b)}} = f(w)$.

$$\frac{e^{-(wn+b)}}{1 + e^{-(wn+b)}} = 1 - \frac{1}{1 + e^{-(wn+b)}}$$

$$\therefore = f(n) f(1-n) (n)$$

$$D_w t = f(n-y) f(n) f(1-n) (n)$$

Similarly we also get

$$D_b t = f(n-y) f(n) f(1-n)$$

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Q3a.

$\frac{\partial L}{\partial h_{t-k}}$ for an RNN.

Let assume a 4 layer RNN S_1, S_2, S_3, S_4 at $t=k$

$$\therefore \frac{\partial L}{\partial h_{t-k}} = \sum_{t=1}^T \frac{\partial L}{\partial S_4} \frac{\partial S_4}{\partial h_{t-k}} \quad \text{--- (1)}$$

for calculation purposes $\frac{\partial L}{\partial h_{t-k}} = \frac{\partial W}{\partial h_{t-k}}$

Now, solve

$$\frac{\partial S_4}{\partial h_{t-k}}$$

In such equation there are always explicit & implicit functions.

Explicit function = $\frac{\partial S_4}{\partial W}$ treat other as constant

Implicit: summation of $S_1 \dots S_4$.

$$\frac{\partial S_4}{\partial W} = \underbrace{\frac{\partial^+ S_4}{\partial W}}_{\text{Explicit}} + \underbrace{\frac{\partial S_4}{\partial S_3} \frac{\partial S_3}{\partial W}}_{\text{Implicit}} \rightarrow \text{can be further expanded in the similar form}$$

$$\frac{\partial S_4}{\partial W} = \frac{\partial^+ S_4}{\partial W} + \frac{\partial S_4}{\partial S_3} \left[\frac{\partial^+ S_3}{\partial W} + \frac{\partial S_3}{\partial S_2} \frac{\partial S_2}{\partial W} \right]$$

$$= \frac{\partial^+ S_4}{\partial W} + \frac{\partial S_4}{\partial S_3} \frac{\partial^+ S_3}{\partial W} + \frac{\partial S_4}{\partial S_3} \frac{\partial S_3}{\partial S_2} \left[\frac{\partial^+ S_2}{\partial W} + \frac{\partial S_2}{\partial S_1} \frac{\partial S_1}{\partial W} \right]$$

$$= \frac{\partial^+ S_4}{\partial W} + \frac{\partial S_4}{\partial S_3} \frac{\partial^+ S_3}{\partial W} + \frac{\partial S_4}{\partial S_3} \frac{\partial S_3}{\partial S_2} \frac{\partial^+ S_2}{\partial W}$$

$$+ \frac{\partial S_4}{\partial S_3} \frac{\partial S_3}{\partial S_2} \frac{\partial S_2}{\partial S_1} \left[\frac{\partial^+ S_1}{\partial W} \right]$$

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after performing short circuit

$$\frac{\delta S_4}{\delta W} = \frac{\delta^+ S_4}{\delta W}$$

$$= \frac{\delta S_4}{\delta S_4} \frac{\delta^+ S_4}{\delta W} + \frac{\delta S_4}{\delta S_3} \frac{\delta^+ S_3}{\delta W} + \frac{\delta S_4}{\delta S_2} \frac{\delta^+ S_2}{\delta W} + \frac{\delta S_4}{\delta S_1} \frac{\delta^+ S_1}{\delta W}$$

$$\therefore \frac{\delta S_4}{\delta W} = \sum_{k=1}^T \frac{\delta S_4}{\delta S_k} \frac{\delta^+ S_k}{\delta W}$$

Now substituting in (v)

$$\frac{\delta L_4}{\delta W} = \frac{\delta L_4}{\delta S_4} \sum_{k=1}^T \frac{\delta S_4}{\delta S_k} \frac{\delta^+ S_k}{\delta W}$$

The final derivation.

$$\frac{\delta L_t}{\delta W} = \frac{\delta L_t}{\delta S_t} \sum_{k=1}^T \frac{\delta S_t}{\delta S_k} \frac{\delta^+ S_k}{\delta W}$$

- For a longer length of memory due to the vanishing gradient we tend to lose information

- In LSTM, we use 3 gates input, output & forget to overcome the issue

- We perform 3 action

(v) Selective ^{writer} Read:- Applying a sigmoid function



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write
we only read the necessary information from the previous state. This helping us to have longer duration memory.

② Selective read:- We perform this for the input at timestep t where we read only the important function and discard the not required one. for stop words etc

③ selective forget:- We get the input as selective read and selective write. We perform forget operation that passes on the required output to $t+1$.

- These operation thus preserves the important information & discards the not necessary information, unlike RNN.

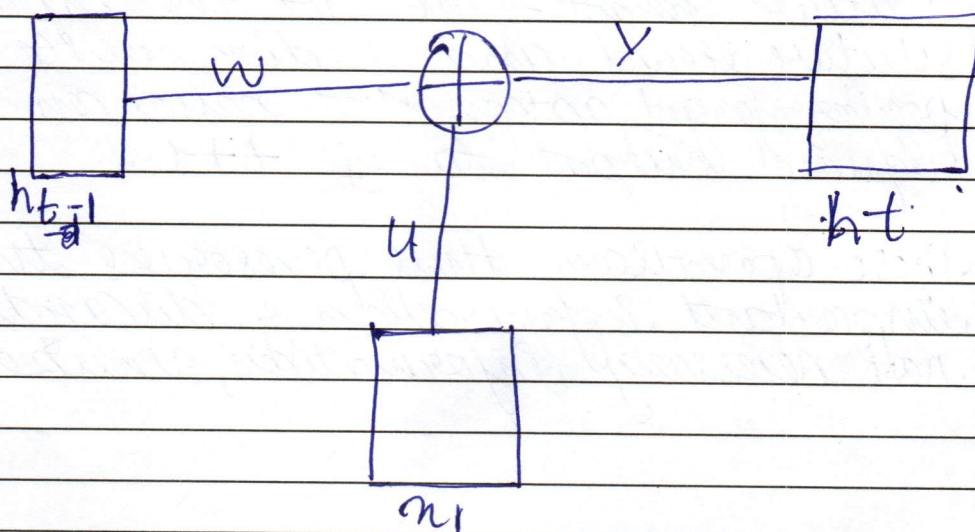
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Q3b.

In GRU we perform 2 gates where we combine 2 gates to update & reset.

$$u_t = \sigma(w_h h_{t-1} + u_n x_t + b_u)$$

$$r_t = \sigma(w_r h_{t-1} + u_r x_t + b_r)$$



~~$$h_t = (1 - z_t) (h_{t-1}) + (z_t) (h_t)$$~~

$$z_t = \tanh((r_t h_{t-1}) w + u(n_t) + b_z)$$

$$h_t = (1 - z_t) (h_{t-1}) + (z_t) (u_t)$$

Since we have used a sigmoid function.
∴ either the state will be updated
or new information will be
added

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- Since we 2 gates where we merge input & forget used in RNN.

∴ The parameters used in GRU are lesser than the parameters used in LSTM.

- This will help the GRU to have descent memory and faster computation. while LSTM will have more memory but slower computation.

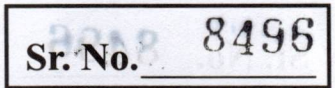
Real life Application.

- GRU:- It can be used language translation, for eg: of Google Translate. As it can store content as well as the real time translation.
- LSTM:- It can be used in financial modeling or grammar correction like Grammarly. As it can store content for a larger time and used in places where accuracy is preferred over computation time.

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Q1b.

- Overfitting occurs when the model is too high as it learns the dataset including the noise too well.
- In such case, the model will give great output at the ~~testing~~ dataset as it has learned it very well. → training
- But it will struggle at testing dataset as during the training phase it failed to identify patterns and thus doesn't perform good.
- That is why we require a bias-variance tradeoff for the same to make sure the model isn't underfit or overfitted.
- Its effect on generalization is sparse as it cannot identify patterns in the accurate format causing ~~caused~~ noisy and inconsistent outputs.
- Thus we can use a validation dataset to ensure that model is trained at the accurate measures to get the best and most accurate output.



Sr. No. 8496

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